

Ram Sharan Chaulagain

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RESEARCH INTERESTS

Design and evaluation of large-scale interconnection networks, network topology and routing algorithms, high-performance computing, software-defined networking, network performance modeling and simulation, distributed systems, and optimizing network performance for emerging AI/ML workloads.

EDUCATION

Ph.D., Computer Science

Expected: Summer 2025

Florida State University, Tallahassee, FL

Advisor: Dr. Xin Yuan

Dissertation: Design and Evaluation of Adaptive Routing for the Next Generation of Interconnection Networks.

M.S., Computer Science

Jan 2025

Florida State University, Tallahassee, FL

Bachelor's in Computer Engineering

January 2017

Tribhuvan University, Nepal

Advisor: Dr. Subarna Shakya

Undergraduate Thesis: Distributed High Performance Computing using JAVA

RESEARCH EXPERIENCE

Research Assistant

Aug, 2018 – Present

Department of Computer Science, Florida State University, Tallahassee, FL

- Analyzed communication characteristics of HPC (Lammps, Nekbone, Milc) and Deep Learning (Cosmoflow) applications. Developed SDN-based adaptive routing schemes tailored to these communication patterns for the Dragonfly interconnect fabric, which avoids congested paths yielding higher throughput.
- Enhanced UGAL routing scheme improving the accuracy of the latency estimation for UGAL with local information showcasing proactive mitigation of the potential network congestion with imbalanced network traffic.
- Implemented collective communications, routing, topologies, and other network parameters to study HPC interconnect performance on the Booksim, CODES, and SST (interconnect simulators).
- Simulated large-scale interconnect systems to identify bottlenecks using various synthetic and application workloads on supercomputers like PSC's Bridges, ALCF's Polaris, and Theta.

Research Intern

May 2020 – Aug 2020

Argonne National Lab, Chicago, IL

- Designed a deep-learning-based routing algorithm matching existing theoretical routing (UGAL-G) with 6% improvement over the existing state-of-the-art UGAL-L routing algorithm for adversarial traffic pattern and published a research poster in Super Computing Conference 2020.

Research Intern

May 2019 – Aug 2019

Argonne National Lab, Chicago, IL

- Developed a counter-based method using local router information for early detection of global link congestion before backpressure and researched congestion mitigation in Dragonfly topology using the CODES simulator.
- Contributed to a CODES interconnect simulator, enhancing the dragonfly-link-stat instrumentation file to log terminal-received packet latency.

TEACHING & OUTREACH

Graduate Teaching Assistant

Fall 2022, Spring 2023

Florida State University, Tallahassee, FL

- **CEN 4020 Software Engineering:** Grading and TA help-desk hours.
- **CGS 2100 Microcomputer Applications (Online):** Grading and TA help-desk hours.
- **CGS 2060 Computer Fluency (Online):** Grading and TA help-desk hours.

Graduate Assistant

Aug 2018 – Jan 2022

College of Education, Florida State University, Tallahassee, FL

- **Virtual Agents Design:** Designed pedagogical, conversational, and interactive virtual agents for a VR simulation-based learning environment, and developed a scalable framework of core design elements for personalized instruction.
- **ML Model Development:** Designed and developed a Bayesian-network classification and prediction model in Unity based 3D game to assess math competency of the players (Middle school students) based on their math competency, achieving 85% accuracy.

PROGRAMMING SKILLS

Programming Languages: C, C++, Python

Parallel Programming Frameworks: POSIX Threads, OpenMP, MPI, CUDA, NCCL

Infrastructure Tools: Git, MySQL, Docker, Terraform

Others: PyTorch, TensorFlow, AWS, Linux scripting, Microcontroller programming, L^AT_EX, Netica, Linden Scripting

HONORS, AWARDS & SERVICE

Outstanding Research Assistant Award (2024): Department of Computer Science, Florida State University

Best Research Presentation Award, CS-Expo'25: Department of Computer Science, Florida State University

Technical Program Committee Member: IEEE Hot Interconnects 2025

REFERENCES

- **Dr. Xin Yuan:** Professor, Department of Computer Science, Florida State University
Email: xyuan@cs.fsu.edu; Phone: +1-850-728-4104
Address: Department of Computer Science, Florida State University, Tallahassee, FL 32306
(PhD Advisor)
- **Dr. Fengfeng Ke:** Professor, College of Education, University of Maryland
Email: fke@umd.edu; Phone: +1-850-536-6818
Address: III7F, Benjamin Building
(Dissertation Committee Member)
- **Sudheer Chunduri:** Software Engineer, Google
Email: sudheerc.atwork@gmail.com; Phone: +1-630-666-8767
Address: Google, Kirkland, WA
(Mentor during Argonne National Laboratory)

PUBLICATIONS

1. **Chaulagain RS**, Mondol T, Bhowmik S, Yuan X. "*Exploiting Software-Defined Networking Technology for Improving UGAL Routing in Dragonfly Networks*" The 25th IEEE/ACM International Symposium on Cluster, Cloud and Internet Computing (CCGRID 2025). 2025 (Accepted)
2. **Chaulagain RS**, Yuan X. "*Enhanced UGAL Routing Schemes for Dragonfly Networks*." ACM International Conference on Supercomputing. 2024 May.
3. **Chaulagain RS**, Yuan X. "*Exploring Software-Defined Networking for Routing in Dragonfly Topology*" ACM/IEEE SC tech poster. 2024 Nov.
4. **Chaulagain RS**, Liza FT, Chunduri S, Yuan X, Lang M. "*Achieving the Performance of Global Adaptive Routing using Local Information on Dragonfly through Deep Learning*." ACM/IEEE SC tech poster. 2020 Nov.
5. West L, Ke F, Dai CP, Kim BJ, **Chaulagain RS**, Mondol TC, Xu J, Kuba R, Jung S. "*Effects of level trajectory on mathematical gameplay*." In Proceedings of the 16th International Conference of the Learning Sciences (ICLS) 2022 2022 Jan.
6. Ke F, Dai Z, Dai CP, Pachman M, **Chaulagain RS**, Yuan X. "*Designing Virtual Agents for Simulation-Based Learning in Virtual Reality*." In Cognitive and Affective Perspectives on Immersive Technology in Education 2020 (pp. 151-170). IGI Global.
7. **Chaulagain RS**, Pandey S, Basnet SR, Shakya S. "*Cloud based web scraping for big data applications*." In 2017 IEEE International Conference on Smart Cloud (SmartCloud) 2017 Nov 3 (pp. 138-143). IEEE.

8. Basnet SR, **Chaulagain RS**, Pandey S, Shakya S. "*Distributed high performance computing in OpenStack cloud over SDN infrastructure.*" 2017 IEEE International Conference on Smart Cloud (SmartCloud) 2017 Nov 3 (pp. 144-148). IEEE.
9. Shakya S, **Chaulagain RS**, Pandey S, Gyawali P. "*Distributed high performance computing using JAVA.*" 2017 International Conference on Computing, Communication and Automation (ICCCA) 2017 May 5 (pp. 742-747). IEEE.

PUBLICATIONS UNDER REVIEW

10. **Chaulagain RS**, Yuan X. "*Improve latency estimation for UGAL routing in Dragonfly Networks*" IEEE Transactions on Parallel and Distributed Systems, Submitted May 2025.

POSTER AND PRESENTATIONS

1. **Chaulagain RS** (Advisor: X. Yuan) "*Exploiting Software-Defined Networking Technology for Improving UGAL Routing in Dragonfly Networks*" Research Presentation at the annual CS Expo, Florida State University, Tallahassee, FL, April 2025.
2. **Chaulagain RS** (Advisor: X Yuan) "*Enhanced UGAL Routing Schemes for Dragonfly Networks*" CS-FSU Department of Computer Science Student Seminar series. Nov 2024
3. **Chaulagain RS** (Advisor: X Yuan) "*Exploring Software-Defined Networking for Routing in Dragonfly Topology*" ACM/IEEE SC tech poster. 2024 Nov.
4. **Chaulagain RS** (Advisor: Sudheer Chunduri) "*Achieving the Performance of Global Adaptive Routing using Local Information on Dragonfly through Deep Learning.*" ACM/IEEE SC tech poster. 2020 Nov.
5. **Chaulagain RS** (Advisor: Sudheer Chunduri) "*Simulation Studies to Categorize Interconnect Congestion Signatures on Dragonfly Topologies*" CODES call: Technical Presentation at the Argonne Leadership and Computing Facility, July 2019